Sahil Khan

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Education

Indian Institute of Technology (ISM), Dhanbad **July 2024 – May 2026**

Master's in Mathematics and Computing

- **Relevant Coursework:** Data Structures and Algorithms, Data Mining, Big Data, DBMS

Skills

Programming: Python, SQL
Machine Learning: Supervised & Unsupervised Learning, Neural Networks, Feature Engineering, Model Evaluation, Cross-Validation,
LLM & Generative AI: RAG Pipelines, Prompt Engineering, Fine-Tuning (LoRA, PEFT), Embeddings, Vector Search (FAISS, Chroma), LLM APIs (OpenAI, Claude), LangChain, LangGraph, MCP Servers, LLM Evals
Agentic AI Systems: Multi-Agent Architectures (CrewAI), Tool/Function Calling, Workflow Orchestration, Intent Routing, Stateful Agents, Human-in-the-Loop (HITL)
MLOps & Deployment: FastAPI, AWS, Docker, Git, DVC, CI/CD, Model Deployment, Monitoring, MLflow

Experience

AI Engineer Intern **May 2025 – July 2025**

Zetheta Algorithms Pvt. Ltd. *Tools Used: Python, scikit-learn, NLP, Pandas*

- Developed an **AI-powered intern–employer matchmaking platform**, transforming unstructured resumes into standardized candidate profiles to enable internship matching for 1,000+ students.
- Built an automated resume parsing pipeline using NER and text preprocessing to extract skills, education, and experience from PDFs and text files.
- Designed a **job readiness scoring engine (0–100)** using weighted evaluation of skill match, education relevance, and project experience to quantify candidate suitability.
- Applied **TF-IDF and cosine similarity** to match candidate profiles with job requirements, improving screening efficiency and role relevance.

Projects

Emotion Analytics from Facial Data (FER-2013 Dataset) | ***Python, TensorFlow, Keras, OpenCV*** | GitHub

- Performed EDA, visualized class distribution, and addressed imbalance using data augmentation and class weighting to improve model robustness.
- Built and compared CNN, VGG16, and ResNet50V2 models, achieving 73% accuracy with detailed precision, recall, and F1-score reporting across seven emotion categories.
- Developed a real-time emotion detection application demonstrating business use cases in sentiment analysis, telemedicine, and e-learning analytics.

Sahil Voice Agent (Agentic AI) | *LangGraph, LangChain, FAISS, GPT-4o, Whisper* | **Demo**

- Built an **agentic RAG system** using LangGraph (ReAct) with conditional retrieval, enabling context-aware responses across multiple knowledge sources.
- Implemented **intelligent query routing** with dynamic tool usage (Tavily) and selective web search for real-time and contextual queries.
- Designed a **stateful conversational pipeline** with memory optimization and FAISS retrieval to improve multi-turn coherence and reduce token usage.
- Developed a **real-time voice AI interface** integrating Whisper (STT) and GPT-4o TTS for low-latency, speech-to-speech interaction.

Achievements

- Qualified **IIT JAM 2024** in Mathematics with **AIR 617 And CUET PG 2024 in Mathematics**.
- Attended **GEN AI Workshops** at **IIT (ISM) Dhanbad** (2026).